Geometry & Coordinate Geometry

Distance and midpoint

1 Find the distance between $(1, 2)$ and $(4, 6)$.

2 Find the midpoint of $(2, -3)$ and $(8, 5)$.

3 Find the distance between $(-2, 1)$ and $(3, -11)$.

4 Find the midpoint of $(-4, 7)$ and $(6, -3)$.

Gradient (slope)

5 Find the gradient of the line through $(1, 2)$ and $(5, 10)$.

6 Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.

7 Find the gradient of the line through $(0, 0)$ and $(6, 9)$.

8 State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.

Equations of lines

9 Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.

10 Find the equation of the line through $(0, 4)$ and $(2, 10)$.

11 Find the x- and y-intercepts of the line $2x + 3y = 12$.

12 Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.

Parallel and perpendicular lines

13 State the gradient of a line perpendicular to a line with gradient 5.

14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

The equation of a circle

16 Write the equation of a circle centered at the origin with radius 6.

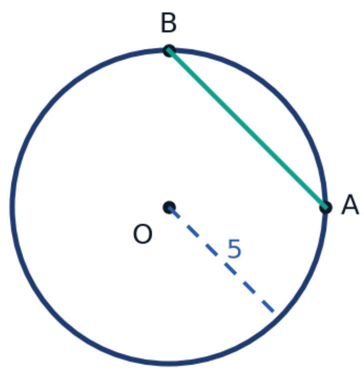
17 Write the equation of a circle centered at $(2, -3)$ with radius 5.

18 Find the radius and center of the circle $(x-1)^2+(y+4)^2=49$.

19 Find the equation of the circle centered at the origin that passes through $(3, 4)$.

Circles: chords and arcs

20 Points A and B lie on a circle of radius 5 centered at O, with central angle $\angle AOB = 90^\circ$. Find the length of chord AB, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.



21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.

22 Find the circumference of a circle with radius 9 cm, in terms of π.

Polygon angle sums

23 Find the sum of the interior angles of a hexagon (6 sides).

24 Find the measure of each interior angle of a regular pentagon.

25 A polygon has an interior angle sum of 1440°. Find the number of sides.

Perimeter and area

26 A rectangle has length 12 and width 7. Find its perimeter and area.

27 A right triangle has legs 6 and 8. Find its area and hypotenuse.

28 A regular hexagon has side length 4, with area $A = \dfrac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

Coordinate geometry: identifying shapes

29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.

30 Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.

31 Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.

The perpendicular bisector of a segment

32 Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.

33 Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.

34 A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.

Word problems: geometry on a coordinate grid

35 This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.

a Find the straight-line distance between the two gates.

b A fountain is placed at the midpoint between the two gates. Find its coordinates.

36 This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.

a Write the equation of the circle covered by the sprinkler.

b Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region.

MCQ — Distance and midpoint

37 Find the distance between $(2, 3)$ and $(6, 6)$.

a. 5

b. 25

c. $\sqrt{7}$

d. 7

38 Find the midpoint of $(3, -4)$ and $(9, 6)$.

a. $(6, 1)$

b. $(6, 2)$

c. $(12, 2)$

d. $(3, 10)$

39 Find the distance between $(-3, 2)$ and $(5, -13)$.

a. 17

b. 15

c. 19

d. 289

40 Find the midpoint of $(-5, 8)$ and $(7, -2)$.

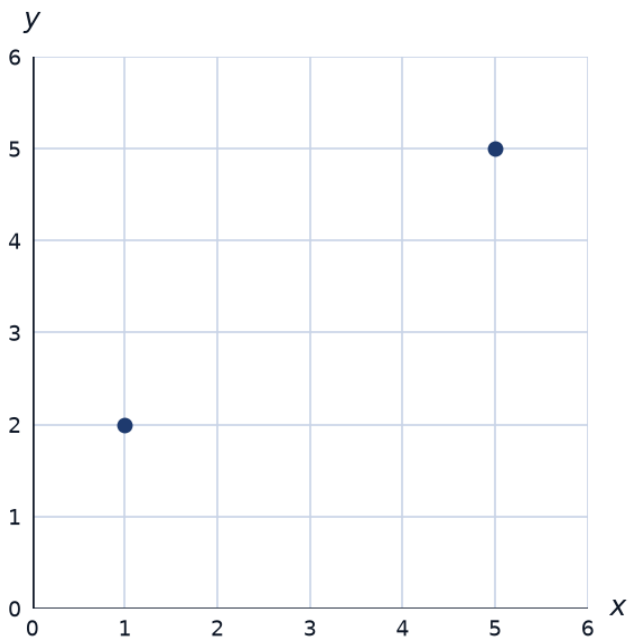
a. $(6, 1)$

b. $(1, 6)$

c. $(2, 6)$

d. $(1, 3)$

41 Find the distance between the two points shown.



a. $\sqrt{5}$

b. 4

c. 3

d. 5

42 Find the midpoint of the segment joining the two points shown.

a. $(4, 3)$

b. $(2, 6)$

c. $(2, 3)$

d. $(8, 6)$

MCQ — Gradient (slope)

43 Find the gradient of the line through $(2, 3)$ and $(6, 11)$.

a. 8

b. 0.5

c. 4

d. 2

44 Find the gradient of the line through $(-2, 5)$ and $(3, -10)$.

a. $-\frac{1}{3}$

b. -15

c. -3

d. 3

45 Find the gradient of the line through $(0, 0)$ and $(8, 12)$.

a. 1.5

b. $\frac{8}{12}$

c. 96

d. $\frac{2}{3}$

46 State whether the line through $(2, 3)$ and $(2, 9)$ is horizontal, vertical, or neither.

a. Cannot be determined

b. Neither

c. Horizontal

d. Vertical

MCQ — Equations of lines

47 Find the equation of the line with gradient 3 passing through $(2, 4)$, in the form $y = mx + c$.

a. $y = 3x + 4$

b. $y = 3x - 2$

c. $y = 3x + 2$

d. $y = 2x - 3$

48 Find the equation of the line through $(0, 5)$ and $(3, 14)$.

a. $y = 5x + 3$

b. $y = 9x + 5$

c. $y = 3x + 5$

d. $y = 3x - 5$

49 Find the x- and y-intercepts of the line $3x + 4y = 24$.

a. $x = 8$, $y = 6$

b. $x = 6$, $y = 8$

c. $x = 24$, $y = 24$

d. $x = 4$, $y = 3$

50 Write the equation of the line parallel to $y = 5x - 2$ passing through $(1, 8)$.

a. $y = 5x + 3$

b. $y = -\frac{1}{5}x + 3$

c. $y = 5x - 3$

d. $y = 5x + 8$

51 Find the equation of the line through $(1, 3)$ and $(4, 12)$.

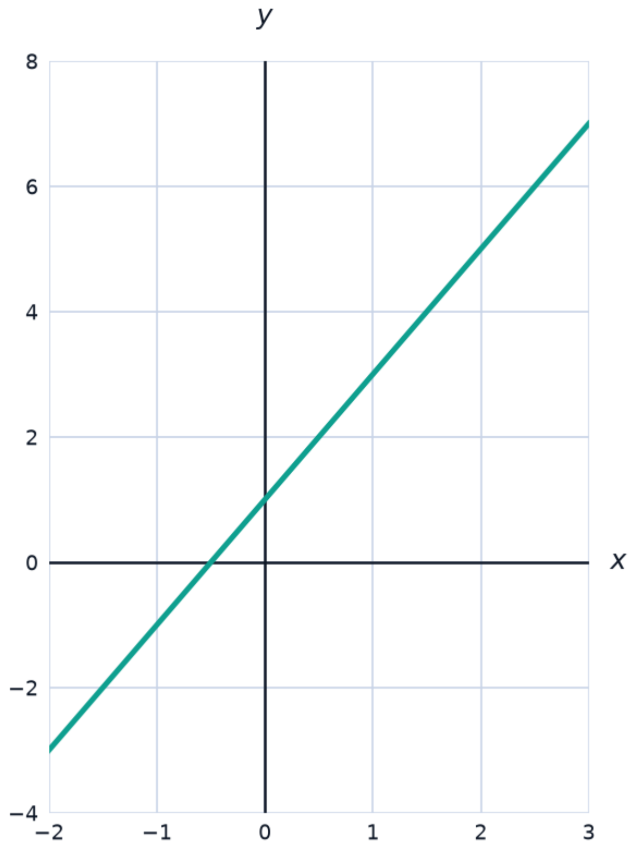
a. $y = 3x + 3$

b. $y = x + 3$

c. $y = 3x - 3$

d. $y = 3x$

52 The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.



a. $y = \frac{1}{2}x + 1$

b. $y = 2x + 1$

c. $y = x + 1$

d. $y = 2x - 1$

MCQ — Parallel and perpendicular lines

53 State the gradient of a line perpendicular to a line with gradient 4.

a. $-\frac{1}{4}$

b. 4

c. $\frac{1}{4}$

d. -4

54 Determine whether the lines $y = 3x + 2$ and $y = 3x - 5$ are parallel, perpendicular, or neither.

a. The same line

b. Neither

c. Perpendicular

d. Parallel

55 Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through $(4, 5)$.

a. $y = 4x - 5$

b. $y = 4x + 11$

c. $y = -\frac{1}{4}x + 6$

d. $y = 4x - 11$

56 Determine whether the lines $y = 2x + 1$ and $y = -0.5x + 4$ are parallel, perpendicular, or neither.

a. The same line

b. Parallel

c. Neither

d. Perpendicular

MCQ — The equation of a circle

57 Write the equation of a circle centered at the origin with radius 8.

a. $x^2 + y^2 = 64$

b. $x^2 + y^2 = 16$

c. $x^2 + y^2 = 8$

d. $(x - 8)^2 + (y - 8)^2 = 64$

58 Write the equation of a circle centered at $(3,-2)$ with radius 7.

a. $(x-3)^2-(y+2)^2=49$

b. $(x-3)^2+(y+2)^2=49$

c. $(x-3)^2+(y+2)^2=7$

d. $(x+3)^2+(y-2)^2=49$

59 Find the radius and center of the circle $(x+2)^2+(y-5)^2=81$.

a. Center $(2,-5)$, radius 9

b. Center $(-2,5)$, radius 9

c. Center $(-2,5)$, radius 81

d. Center $(2,5)$, radius 9

60 Find the equation of the circle centered at the origin that passes through $(6,8)$.

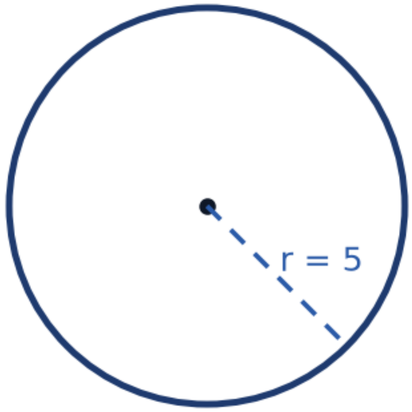
a. $x^2+y^2=100$

b. $x^2+y^2=10$

c. $x^2+y^2=196$

d. $x^2+y^2=48$

61 Write the equation of the circle shown, centered at the origin.



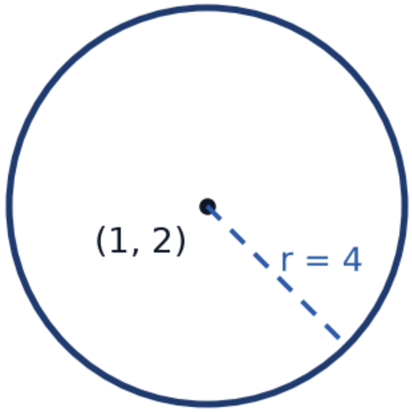
a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$

62 The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.



a. $(x-1)^2+(y-2)^2=8$

b. $(x+1)^2+(y+2)^2=16$

c. $(x-1)^2+(y-2)^2=16$

d. $(x-1)^2+(y-2)^2=4$

MCQ — Circles: chords and arcs

63 Points A and B lie on a circle of radius 8 centered at O, with central angle $\angle AOB = 60^\circ$. Find the arc length AB, in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{4\pi}{3}$

c. 16π

d. $\frac{8\pi}{3}$

64 A chord of a circle with radius 13 cm is 24 cm long. Find the distance from the center to the chord.

a. 7

b. 1

c. 12

d. 5

65 Find the circumference of a circle with radius 11 cm, in terms of π.

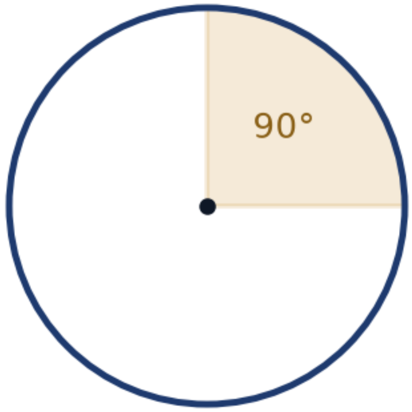
a. 121π

b. 44π

c. 22π

d. 11π

66 Find the arc length of the 90° sector shown, in terms of π.



a. 12π

b. 3π

c. 1.5π

d. 6π

MCQ — Polygon angle sums

67 Find the sum of the interior angles of an octagon (8 sides).

a. 720°

b. $1{,}440^\circ$

c. 900°

d. $1{,}080^\circ$

68 Find the measure of each interior angle of a regular hexagon.

a. $140°$

b. $135°$

c. $108°$

d. $120°$

69 A polygon has an interior angle sum of $1{,}620°$. Find the number of sides.

a. 10

b. 12

c. 9

d. 11

70 Find the sum of the interior angles of a decagon (10 sides).

a. $1{,}800°$

b. $1{,}260°$

c. $1{,}080°$

d. $1{,}440°$

MCQ — Perimeter and area

71 A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

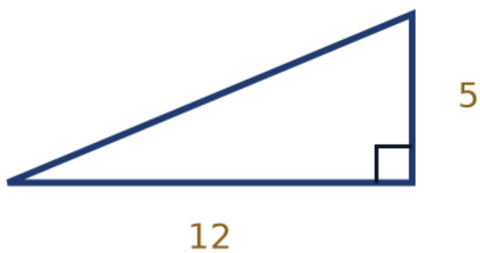
a. $P=135$, $A=48$

b. $P=48$, $A=135$

c. $P=24$, $A=135$

d. $P=48$, $A=24$

72 A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.



a. $A=60$, hyp $=13$

b. $A=30$, hyp $=17$

c. $A=30$, hyp $=13$

d. $A=30$, hyp $=12$

73 A regular hexagon has side length 6, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

a. $36\sqrt{3}$

b. $108\sqrt{3}$

c. $54\sqrt{3}$

d. $18\sqrt{3}$

74 A regular pentagon has side length 8, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.

a. 110

b. 124

c. 96

d. 14

MCQ — Coordinate geometry: identifying shapes

75 Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

a. No

b. Cannot be determined

c. Yes

d. Only if it is also isosceles

76 Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.

a. Only if it is also equilateral

b. No

c. Cannot be determined

d. Yes

77 Determine whether the triangle with vertices $A(2, 3)$, $B(6, 3)$, $C(4, 7)$ is isosceles.

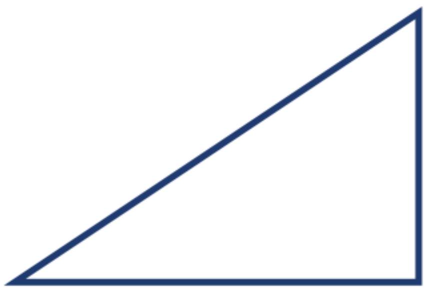
a. No

b. Only if it is also a right triangle

c. Yes

d. Cannot be determined

78 Is the triangle shown a right triangle?



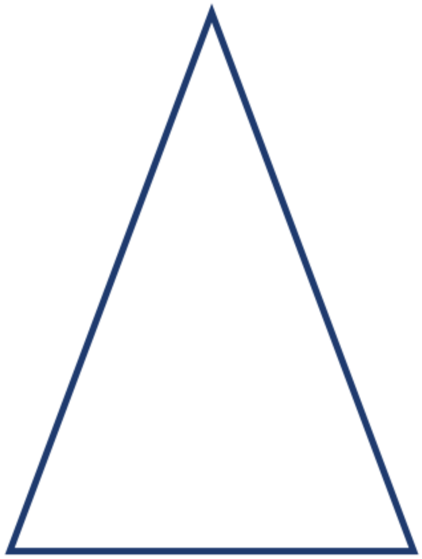
a. Only if it is also isosceles

b. Cannot be determined

c. No

d. Yes

79 Is the triangle shown isosceles, scalene, or equilateral?



a. Isosceles

b. Cannot be determined

c. Equilateral

d. Scalene

MCQ — The perpendicular bisector of a segment

80 Find the equation of the perpendicular bisector of the segment joining $(1, 4)$ and $(7, 10)$.

a. $y = -x + 11$

b. $y = x - 11$

c. $y = x + 11$

d. $y = -x + 7$

81 A line segment has endpoints $(2,0)$ and $(2,8)$. Find the equation of its perpendicular bisector.

a. $x=4$

b. $y=4$

c. $y=2$

d. $x=2$

82 A line segment has endpoints $(3,6)$ and $(9,6)$. Find the equation of its perpendicular bisector.

a. $x=3$

b. $y=6$

c. $y=9$

d. $x=6$

83 Find the equation of the perpendicular bisector of the segment joining $(-2,1)$ and $(4,9)$.

a. $y=-\frac{3}{4}x+\frac{23}{4}$

b. $y=-\frac{3}{4}x+5$

c. $y=-\frac{4}{3}x+\frac{23}{4}$

d. $y=\frac{3}{4}x+\frac{23}{4}$

MCQ — Word problems: geometry on a coordinate grid

84 A city park is designed on a coordinate grid (in meters). Two gates are located at $(10,20)$ and $(50,60)$. Find the distance between the gates, to the nearest meter.

a. 113

b. 40

c. 57

d. 80

85 A circular irrigation sprinkler is centered at $(8, 8)$ on a farm's coordinate grid (in meters) and has a radius of 15 meters. Write the equation of the circle it waters.

a. $(x+8)^2 + (y+8)^2 = 225$

b. $(x-8)^2 + (y-8)^2 = 15$

c. $(x-8)^2 + (y-8)^2 = 225$

d. $(x-8)^2 + (y-8)^2 = 30$

86 A surveyor marks two points on a coordinate map (in km): a base camp at $(3, 4)$ and a summit at $(15, 4)$. Find the midpoint, to be used as a resupply station.

a. $(6, 4)$

b. $(9, 8)$

c. $(9, 4)$

d. $(18, 8)$